
WorldEpisode: Auditing Silent State Drift in Robot-Learning Datasets

Anonymous Author(s)

Affiliation

Address

email

Abstract

Robot-learning datasets can be locally valid and behaviorally wrong: every file loads, but a command is replayed at the moment it was issued rather than when the motor executed it, the same reconstructed scene appears in both training and testing, or an asset’s collision role no longer matches the state under evaluation. We call this *silent state drift*. This paper presents *WorldEpisode*, an executable state-integrity contract for robot-learning datasets: five machine-checkable invariants per episode (persistent identity, frame and clock mappings, representation roles, temporal events, provenance) plus the two corpus-level obligations that carry the headline results, lineage-disjoint evaluation splits and loss-explicit conversion.

On a public SO-101 LeRobot release, a random episode split overlaps every tested task–scene proxy lineage. Across 5 optimization seeds, mean episode nRMSE is 0.183 [0.171, 0.196]; holding out 2 task–scene groups reduces measured proxy-lineage overlap to 0.000 and increases nRMSE to 0.381 [0.364, 0.399], a 0.197 increase [0.177, 0.219]. Because task identity is part of the lineage key, this result does not isolate scene leakage from task shift. A 3-frame lag frozen on 320 trajectories reduces held-out pooled action/state telemetry RMSE from 4.845 to 1.934 source position units on 80 trajectories; the mean paired episode improvement is 3.087 [2.890, 3.286]. This is a telemetry proxy spanning 1 controller configuration and without motor-effective timestamps. A complete-shard round-trip audit over 271 public LeRobot episodes from 3 datasets preserves 43,601 paired state/action samples with maximum numerical error 0.0 while reporting source-absent semantics as conversion loss; the validator detects all 14 injected fault classes. We claim neither simulator-independent physics nor benchmark inflation without faithful policy reruns.

1 Introduction

Robot-learning data can be structurally valid yet behaviorally wrong. A tensor can have the expected shape but an undeclared unit; a command timestamp can be mistaken for motor-effective time; an asset can retain its appearance while losing its collision role; or an episode-level split can put related world captures in both training and evaluation. Each artifact still loads, so ordinary schema and container checks miss the failure. We call this class *silent state drift*: the stored observation, action, asset, or split no longer denotes the physical state assumed by training, replay, or evaluation.

Robot learning is a demanding test case because it combines asynchronous control, multi-camera recordings, reconstructed scenes, simulator exports, and policy evaluation. LeRobot integrates collection, datasets, policies, and deployment [4]; LeRobotDataset v3 stores state, action, timestamps, video, and episode indexes [7]. Rerun and NCore cover multimodal logs and pose conventions, while OpenUSD and glTF describe worlds and assets [2, 15, 19]. These containers are complementary, but

Table 1: The contract at a glance: each question a valid episode must answer, the graph or obligation that answers it, the executable requirement IDs, and the evidence case that tests it.

Question	Answered by	Requirements	Tested in
Which state revision am I in?	world revision + ordered deltas	WORLD.001, TRACE.001	Case 4
What entities does it contain?	identity graph	ENTITY.001	Case 4
Where and when is it valid?	frame + clock graphs	FRAME.001-2, TIME.001-2, ACTION.001-2	Cases 2, 4
What is each asset safe for?	representation-role graph	REP.001, ASSET.001-2	Cases 3, 4
What actually happened?	temporal-event graph	TRACE.001	Case 4
Where did this come from?	provenance graph	PROV.001, QUALITY.001	Case 4
What was lost in conversion?	obligation: loss-explicit conversion	CONVERT.001	Cases 5, 6
Can the evaluation split leak?	obligation: lineage-safe splits	SPLIT.001	Cases 1, 7
Can this scale past one folder?	dataset manifest + indexes	DATASET.001	App. F

none by itself proves that the same entities, frames, clocks, action semantics, asset roles, and source lineages survive conversion.

Related work. Documenting datasets is not new. Datasheets for Datasets [5] and model cards [14] standardize prose descriptions of provenance and intended use; MLCommons Croissant [1] adds a machine-readable metadata layer for ML-ready datasets; and Open X-Embodiment [17] aggregates robot data under a shared schema. Evaluation leakage is a well-documented driver of the reproducibility crisis in ML-based science [8]. WorldEpisode differs by making cross-container assumptions executable. It binds each episode to an immutable world revision and closed identity, frame/clock, representation-role, event, and provenance graphs; a fail-closed validator reports stable requirement IDs, and each conversion emits machine-readable loss. It is a sidecar contract over existing containers, not a replacement format or dataset card.

Contributions.

1. An executable state-integrity contract with five coupled graphs, JSON Schema, semantic validator, conformance fixtures, and fail-closed LeRobot/Rerun preflight.
 2. Loss-explicit LeRobot conversion, including exact round trips over 271 public episodes from 3 datasets and a versioned binding-retention audit.
 3. Empirical audits of a task-scene-proxy split and action timing, with episode-level uncertainty intervals and explicit boundaries for the confounded split, single-controller telemetry, same-trace and contact-rich simulator replay, and controlled proxy experiments.
- Section 2 states the contract, Section 3 makes it executable, Section 4 reports the audits, and Section 6 states their limits.

2 The Contract: Five Graphs and Two Corpus Obligations

A robot-learning episode must answer a practical question: what must be known before it can be safely converted, replayed, synchronized, augmented, or used for evaluation? WorldEpisode answers it with five coupled graphs at the level of a single episode record, two corpus-level obligations that follow from those graphs, and one scale requirement for production datasets. Table 1 maps each question to the structure that answers it, the requirement IDs that make it checkable, and the case in Section 4 that tests it. WorldEpisode specializes these requirements for robot-learning episodes.

2.1 The five coupled graphs

Figure 1 shows the five graphs that make a spatial state replayable and auditable rather than just stored. Making them mutually referential is what turns a silent error into a detectable one: an identity, timing, role, event, or lineage error becomes a broken reference instead of a hidden assumption.

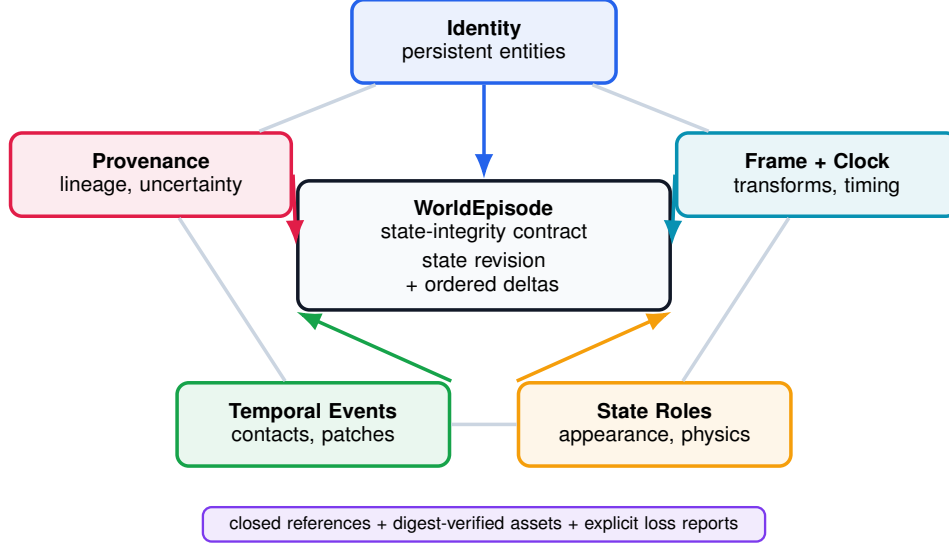


Figure 1: The five-graph model. WorldEpisode makes identity, frames/clocks, state roles, temporal events, and provenance mutually referential so conversion, synchronization, and replay errors become detectable instead of hidden.

69 **World revision and deltas** (WORLD.001, TRACE.001). Each record references one immutable,
70 content-addressed world revision plus an ordered list of deltas; any change to geometry, scale,
71 calibration, collision proxies, physical parameters, or object identity creates a new revision. Content
72 hashes let two datasets prove they refer to the same world without relying on filenames.

73 **Identity graph** (ENTITY.001). Every physical or logical entity receives a persistent identifier,
74 and the same `entity_id` must be used by segmentation masks, Gaussian groups, meshes, collision
75 bodies, simulator actors, language references, and interaction events. If a language command says
76 “pick up the mug,” the perception mask, the simulator object, and the collision proxy must identify
77 the same mug.

78 **Frame and clock graphs** (FRAME.001, TIME.001, ACTION.002). Every spatial value declares
79 its frames, units, transform direction, and validity interval, consistent with and extending ROS
80 conventions [16]; every stream declares a clock domain, timestamps, synchronization mapping, and
81 estimated drift, with the rigor NCore demonstrates for pose graphs [15]. Actions declare command
82 time and effective motor time rather than assuming those are the same instant. This graph turns a
83 suspicious replay from “the policy is bad” into a diagnosable offset, transform-direction, or unit error.

84 **Representation-role graph** (REP.001). One entity can have many representations (splat, mesh,
85 point cloud, collision proxy, mass/inertia record), each declaring a role and a digest-verified asset
86 descriptor. The role field is essential: a splat may be valid for appearance but invalid for collision,
87 while a simplified proxy may be valid for collision but unsuitable for visual training.

88 **Temporal-event graph** (TRACE.001). Records store interaction events — contact, grasp, attach-
89 ment, collision patch, reset, intervention, failure — each referencing entities, time, confidence, and
90 source, turning raw coordinate streams into auditable milestones.

91 **Provenance graph** (PROV.001). Every derived asset identifies its source episodes, sensors, recon-
92 struction algorithm and version, calibration revision, hashes, creator, license, and quality metrics,
93 mapping to existing provenance vocabularies rather than inventing an incompatible one.

94 The full normative field lists are specified in the v0.1 RFC and its JSON Schemas; the paper keeps
95 only what each graph must guarantee.

96 2.2 Two corpus-level obligations, plus scale

97 **Loss-explicit conversion** (CONVERT.001). WorldEpisode reuses existing standards rather than
98 replacing them. USD [2] carries composed simulation worlds, glTF carries Gaussian appearance
99 assets [19], NCore carries capture and pose-graph conventions [15], Rerun carries physical-data
100 recordings with column chunks [18], ROS/MCAP carries logs, and domain-native engines carry
101 their own telemetry. A converter may externalize or approximate fields, but it must emit a machine-
102 readable loss report instead of silently discarding semantics. The standard permits lossy conversion;
103 it forbids silent lossy conversion.

104 **Lineage-safe splits** (SPLIT.001). Record-level random splits are not enough when multiple records
105 share a reconstructed room, map patch, physical object, source capture, calibration, generated asset
106 lineage, or state ancestry — exactly what the provenance graph records. A WorldEpisode split mani-
107 fest can therefore enforce disjointness by constraints such as `world_lineage`, `physical_entity`,
108 `source_capture`, `reconstruction_run`, or `generated_asset_family`.

109 **Scale** (DATASET.001). Large robot corpora require a dataset manifest above the episode records:
110 namespaces, resolver priority, shard catalogs, materialized indexes, split catalogs, and append-only
111 versions. This lets a reader locate all SO-101 pick-place traces for a world-disjoint split by index
112 lookup rather than by crawling storage paths.

113 The initial WorldEpisode profile is intentionally narrow: rigid tabletop manipulation with fixed-
114 base single- or dual-arm robots. That bounded domain keeps conformance executable rather than
115 aspirational. Future profiles can extend this specification to humanoids, locomotion, deformables,
116 fluids, multi-agent game worlds, and autonomous-driving logs.

117 3 The Sidecar Contract, Made Executable

118 WorldEpisode delivers the contract as a sidecar blueprint around existing formats. It does not ask a
119 LeRobot dataset to become a USD scene, or a USD scene to become a policy-training table. Instead,
120 it records the relationships those containers need in order to agree about the same state: the state
121 revision, persistent entities, space/time conventions, transition contract, asset roles, interaction events,
122 provenance, and split lineage. The sidecar is not a mandate to package a global robot corpus as
123 millions of small folders: the single-episode record is the semantic unit, and production datasets add
124 a manifest layer (namespaces, URI resolvers, shard catalogs, materialized indexes, split manifests,
125 append-only snapshots) so that opening a corpus starts from the manifest and its indexes rather than
126 from a recursive storage scan.

127 Native containers continue to store the arrays, video, logs, worlds, and assets for which they were
128 designed. The sidecar stores only the relationships otherwise scattered across loader assumptions,
129 making cross-container identity, timing, role, lineage, and conversion loss checkable.

130 3.1 Runtime adapters

131 WorldEpisode is not tied to a single rendering or physics engine. A runtime becomes a trusted
132 target by implementing an adapter that preserves the same semantic invariants and emits replay
133 evidence, which keeps the integration direction clean: datasets are not rewritten around simulator-
134 native assumptions; simulators compile their scene loading, action interfaces, and replay reports
135 against the sidecar. The adapter contract has three layers:

- 136 • **Invariant interface**: ingests immutable world revisions, frame and clock graphs, entity identity,
137 representation roles, action channels, asset digests, provenance, and quality records.
- 138 • **Asynchronous schema extension**: lets an engine register cloth, deformable, fluid, articulation,
139 material, or procedural-scene extensions without weakening the core episode record.
- 140 • **Deterministic replay accountability**: records runtime identity, version, solver, timestep, actuator
141 parameters, initialization state, command/effective timing, latency, tolerance envelopes, and
142 measured drift.

The current implementation includes tested MuJoCo [20] and Genesis [6] adapters for the same LeRobot control-loop trace and a preregistered two-task primitive contact protocol. The Isaac mapping is emitted but untested, and SAPIEN remains adapter-required. The claim is therefore not simulator-independent physics but runtime-neutral accountability: every simulator reports against the same state contract, so cross-runtime drift becomes measurable rather than hidden inside loader conventions.

3.2 The executable robotics profile

Complete action semantics. An action vector is not portable without its operational contract. Each action channel must declare actuator, control mode, parameterization, reference frame, units, absolute/delta/velocity semantics, limits, command rate, command timestamp, effective timestamp, latency model, interpolation, normalization, gripper semantics, and missing-value policy. A vector such as $[0.01, 0, 0, 0, 0, 0, 1]$ is otherwise ambiguous: it could be a tool-frame delta, base-frame delta, velocity command, normalized policy output, or delayed target increment.

Implementation artifact. We implement the contract as four executable components: a JSON Schema for structural validity, a machine-readable conformance requirement file, an installable semantic validator, and a round-trip experiment runner. The validator maps each diagnostic to a stable requirement identifier such as `TIME.001`, `FRAME.002`, or `ASSET.002`. The experiment runner takes a `WorldEpisode` package, exports binding-native artifacts plus sidecars, reimports them, compares a versioned semantic projection, injects controlled faults, and writes reproducible result artifacts. The projection itself is published as `conformance/projections/uss-core-23.v0.json` and validated against `schemas/semantic-projection-v0.schema.json`. The contract is represented independently in the public schema and validator rather than only in its initial implementation.

One-line preflight. The validator is packaged as `worldepisode`. A blocking preflight can be run from the shell with `worldepisode preflight <dataset-or-manifest>` or from a training script with a single Python call such as `preflight_lerobot(dataset.root).raise_if_failed()`. Native LeRobot folders and Rerun .rrd recordings are recognized directly; without a `WorldEpisode` manifest or sidecar, they fail closed on missing world revision, entity identity, representation role, action timing, frame/clock, split-lineage, and conversion-loss controls. This makes the semantic audit a pre-training check rather than a separate post-hoc documentation exercise. The repository pairs the validator with converters, leakage and replay tools, a clean-room reader, and a runtime adapter report; Section 4 presents what each one measures.

4 Evidence: What the Audit Finds

We test what drift changes (Cases 1–3), what the contract detects or preserves (Cases 4–6), and which controls remain unavailable in 5 public benchmarks (Case 7). All measured data are robotics. Appendix A records claim boundaries and Appendix E records unexecuted gates.

4.1 What drift costs

Case 1: task-scene proxy leakage in evaluation splits (SPLIT.001). We audit 400 teleoperated episodes from ArmnetBench v0.1’s SO-101 LeRobot release [3]. Public metadata does not expose physical scene IDs, so we hash source revision, robot type, task ID/text, and camera schema into 8 *task-scene proxy* lineages. A seeded 80/20 episode split (320/80 episodes) puts every test proxy in training. The comparison split holds out task IDs 3 and 4 (300/100 episodes), yielding 0.000 proxy overlap. *This is also a task-disjoint shift*; it cannot identify the causal effect of scene overlap.

A 2-layer Torch MLP (64 units/layer, AdamW, 12 epochs, 5 seeds) maps state, episode phase, timestamp, and task one-hot features to action. The primary metric, mean episode nRMSE, changes from 0.183 (± 0.001 seed SD; 95% crossed seed-episode bootstrap CI $[0.171, 0.196]$) to 0.381 (± 0.020 ; $[0.364, 0.399]$); the increase is 0.197 $[0.177, 0.219]$. As a secondary diagnostic, the rate below a fixed 0.25 nRMSE threshold is 0.853 (± 0.016 seed SD) versus 0.000 (± 0.000). This threshold was not selected on a validation set and is not task success. Task one-hot features make this probe particularly sensitive to the task shift.

A deterministic ridge policy using only a 3-frame state history has mean episode nRMSE 0.157 [0.145, 0.170] versus 0.255 [0.247, 0.262]; the increase is 0.097 [0.082, 0.112]. Its secondary thresholded rates are 0.925 (74/80; [0.846, 0.965]) versus 0.420 (42/100; [0.328, 0.518]); its independently bootstrapped rate difference is 0.505 [0.390, 0.615]. We use 20,000 resamples. MLP intervals cross optimization seeds with held-out episodes, pair sampled seeds across protocols, and resample the different episode sets independently; ridge intervals resample episodes only. Thus the experiment establishes that the random split is not proxy-lineage-disjoint and that evaluation changes materially under a harder task-scene holdout; it does not establish scene-only leakage, policy rollout impact, or benchmark inflation.

The artifact includes digest-verified split packages and ACT/Diffusion job specifications. A pinned LeRobot 0.6.0 preflight first loads the representative low-dimensional package; 0/2 policy paths complete a training step, and all 2 stop because the package has neither an observation image nor an environment-state feature. We do not relabel joint proprioception to bypass that contract. A subsequent generated plan pins 24 source front-camera files (1.74 GiB) and materializes split-specific video metadata and statistics for all 4 packages. With those inputs, 2/2 pinned LeRobot 0.6.0 ACT/Diffusion paths complete the CUDA smoke optimization step. This closes input compatibility only: checkpoint training, held-out policy metrics, and rollouts remain open (Appendix E).

Case 2: action-timing and replay drift (ACTION.002). LeRobot’s control stack can decouple policy inference from the fixed-rate low-level controller: commands may be queued, chunked, interpolated, or consumed after policy output [4]. We estimate an integer action/state lag from 320 calibration trajectories spanning 8 tasks, freeze it, and test 80 source-episode-disjoint trajectories with 0 source-episode overlap. The selected lag is 3 frames (150 ms). On identical within-episode target rows, zero-delay alignment has pooled RMSE 4.845 source position units and frozen-delay alignment has 1.934. The mean paired episode improvement is 3.087 (95% CI [2.890, 3.286]); all 80/80 held-out trajectories improve.

This is an action/state telemetry-lag proxy, not measured motor latency. The source exposes 1 float32 frame-timestamp column, no motor-effective timestamps, 1 robot type, and 1 controller configuration. Using the derived time shift, timestamp zero-order hold gives 2.021 RMSE whereas linear interpolation gives 1.934, showing that the timestamp comparison is interpolation-sensitive rather than independent queue evidence.

Separately, 1 trajectory has 303 6-joint samples at 30 Hz; its calibration prefix selects 4 frames, or 133 ms, and its held-out suffix changes alignment RMSE from 4.732 to 1.862 degrees. A WorldEpisode contract declares the estimated command/effective semantics, interpolation, and latency model. We execute it through 2 minimal position-servo adapters: MuJoCo changes from 3.425 to 1.563 degrees, while Genesis same-trace replay changes from 3.425 to 1.563. Their naive values (3.4247237 and 3.4247236) are nearly identical because both integrate the same minimal servo model, not because their physics agree. The Isaac adapter is emitted as a ready mapping from the same contract, but is not tested in this environment.

A dependency-free replay-adapter harness covers 3 delay, hold, and queue cases; the contract-aware scheduler has 0.0 maximum RMSE. Separately, a preregistered contact protocol executes a straight push and parallel-jaw capture in MuJoCo and Genesis using the same content-addressed primitive world, sampled actor poses, clock, and 16 initial states per task (32 paired scenarios). Object-trajectory RMSE is 9.01 mm [7.38, 10.65]; frame-aligned contact precision/recall/F1 are 0.995/0.964/ 0.979 (F1 CI [0.968, 0.989]). Grasp-state and task-outcome agreement are 0.997 and 1.000, respectively. Yet final orientation differs by 26.2° overall [13.6, 40.6] and 52.4° on capture [33.0, 72.4]. Thus the result makes contact drift measurable; scripted actors, no hardware ground truth, and this orientation divergence preclude an equivalent-physics claim.

Case 3: controlled real-to-sim contract drift (ACTION.001, REP.001). 2 deterministic proxies isolate an absolute-radian versus delta-degree controller mismatch and an appearance-only Gaussian export that drops collision geometry. Drifted contracts produce 2/2 proxy simulation successes and 0/2 deployment-proxy successes; explicit action and representation-role contracts recover 2/2. This is a mechanism check, not a hardware or RoboSnap rerun.

Figure 2 summarizes the measured changes for the displayed experiment families where explicit state semantics alter the outcome.

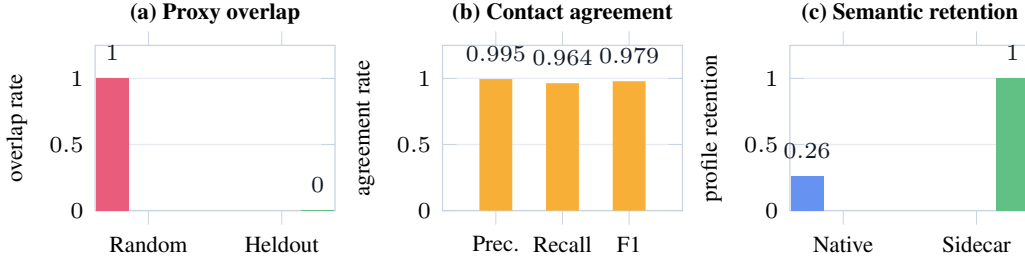


Figure 2: Summary of the displayed audits. Proxy overlap is confounded with task holdout; contact agreement pools 32 paired scenarios; sidecar retention is by construction.

247 4.2 What the contract catches

248 **Case 4: validator fault detection (all requirement IDs).** We inject 14 faults into the baseline
 249 package: missing clock domains, incomplete cross-clock mappings, unknown frames, missing
 250 transform intervals, unknown event entities, missing representation roles, digest errors, action-
 251 contract omissions, non-content-addressed worlds, missing delta sequences, missing provenance, and
 252 missing quality records. The validator detects all 14 expected requirement failures with 1.000 recall
 253 and 0.933 precision. 2 hand-authored independent fixtures also trigger the expected diagnostics. The
 254 single false-positive requirement is useful rather than harmful: removing an action reference frame
 255 triggers both ACTION.001 and the downstream FRAME.001 reference check. The same suite includes
 256 a unit sanity check from lerobot/svla_so101_pickplace: the first mirrored SO-101 frame has
 257 6 joint values with maximum magnitude 98.924 in degrees and 1.727 after radian conversion, so
 258 treating those degree values as radians would violate the action/state unit contract even though the
 259 tensor shape is valid.

260 A pilot natural-source audit records 19 source omissions or evaluation risks across 2 active LeRobot
 261 artifacts plus source-level DROID and BridgeData V2 metadata. These are not maintainer-confirmed
 262 bugs or prevalence estimates. A 4-case preflight regression passes a valid sidecar and rejects an invalid
 263 manifest plus native LeRobot/Rerun artifacts that lack sidecars. An internal reader independent of
 264 the SDK path detects all 18 expected requirement instances across 17 fixtures; this tests artifact
 265 readability, not external adoption.

266 **Case 5: loss-explicit dataset conversion (CONVERT.001).** To test an actual LeRobotDataset
 267 rather than only a capability model, we implement LeRobotDataset v3 $\rightarrow$ WorldEpisode $\rightarrow$
 268 LeRobotDataset v3 on 5-episode batches from 2 pinned public datasets: lerobot/svla_so101_
 269 pickplace [11] and lerobot/pusht [10]. The converter downloads the real Parquet metadata and
 270 data shards, builds a schema-valid WorldEpisode manifest and sidecar, exports a compact LeRobot-
 271 Dataset v3 package, reads it back, and asserts exact equality. These 10 committed episode packages
 272 provide inspectable per-episode evidence.

273 For scale, we convert every episode assigned to one immutable source Parquet shard from each of
 274 3 datasets: the two above and ArmnetBench v0.1 [3]. No episode-level sampling is used. Across
 275 271 episodes, including 2 multi-camera datasets, the round trip preserves 43,601 source-native
 276 paired action/state samples, indices, sample timestamps, and video timestamp ranges with maximum
 277 absolute error 0.0. The 5.2 MiB input produces 17.5 MiB of temporary packages in 48.24 s with
 278 393.3 MiB peak resident memory. Source video payloads are neither downloaded nor converted; only
 279 their stream and timestamp metadata are audited. This is a complete-selected-shard measurement,
 280 not full-corpus throughput.

281 Both audits preserve derived physical-frame records through the sidecar and do not fake missing
 282 semantics. The same 4 source-absent fields—camera extrinsics, robot/world calibration, action units,
 283 and controller latency—are reported as conversion loss rather than guessed.

284 **Case 6: binding retention (CONVERT.001).** 7 pilot bindings are compared against a versioned
 285 23-field projection. Non-reference native retention ranges from 0.174 to 0.391; sidecars recover 1.000
 286 for tested dataset/log/world bindings by serializing the missing fields. This is a round-trip check and
 287 gap inventory, not a universal format ranking; Appendix Table 3 reports the per-binding values.

4.3 Scaling the audit

Case 7: source-level benchmark audit (SPLIT.001). We audit public metadata for Open X-Embodiment/RT-X [17], DROID [9], BridgeData V2 [21], LIBERO [12], and CALVIN [13]. The audit asks whether public metadata exposes lineage splits, content-addressed world revisions, persistent entity identity, action timing contracts, frame/clock graphs, and loss reports. Each has at least one high-severity open control. A DROID-100 proxy rerun over 100 episodes and 32,212 frames scores 0.516 under its corrected proxy split versus 0.496 randomly, so it provides no inflation evidence. The proof gate therefore records 0 famous-benchmark inflation claims.

The same audit protocol also runs above the episode level: a dataset-scale manifest audit and a generated catalog benchmark (Appendix F) make the DATASET.001 scale claim executable on catalog-side evidence only.

5 Positioning

A contribution of falsifiable invariants. The experiments show several ways a result can be wrong while the files are valid: a split can leak world lineage, replay can drift because control timing is underspecified, conversion can discard physical semantics that the target container has no native place to store, and a live virtual or AV state can drift from its authoritative revision. WorldEpisode is useful only because these failures are observable and testable. The sidecar is the mechanism; the scientific claim is that explicit spatial-state semantics change what an embodied-agent result means. A container adapter either preserves those invariants, externalizes them with declared loss, or fails conformance.

A small semantic core with multiple bindings. The bindings are evaluated at very different depths: LeRobotDataset v3 is the only binding exercised with active real-data experiments (conversion, leakage, replay timing); Rerun, NCore, MCAP/ROS, OpenUSD/SimReady, and glTF/SPZ are covered only as binding-retention pilots (Table 3). OpenUSD standardizes how the 3D world is composed; WorldEpisode standardizes how a robot modifies state within that space over time without silent data corruption. The same invariant structure should generalize to a game-engine or AV profile, though only the robotics profile is implemented and evaluated here.

Real-to-sim pipelines and benchmarks are the users, not the competitors. Gaussian-splat and OpenUSD real-to-sim pipelines are the systems that most need a state and episode contract: a reconstructed room can be excellent visual training data and still be unsafe for policy evaluation if collision roles are missing, lineage leaks across splits, or the replayed action vector is interpreted under the wrong hardware contract. The same stance governs benchmark auditing: when public metadata cannot answer whether train and test share a scene lineage, asset family, source capture, or timing convention, WorldEpisode marks the published score as unaudited with respect to that control — and marks it inflated only after the same policy protocol is rerun under the corrected split or timing contract.

Standards alignment. The project aligns with standards bodies rather than antagonizing them. Static 3D robot-map exchange, humanoid dataset lifecycle, OpenLABEL annotation, glTF Gaussian splats, USD composition, and NCore capture conventions are all useful neighbors. WorldEpisode is the executable profile and bridge across those layers.

6 Limitations and Open Gates

The current evidence is deliberately scoped: it exposes hidden failure mechanisms rather than claiming full robot-learning, game-engine, or autonomous-driving generality. The boundary of every empirical claim is tabulated in Appendix A, and each open gate’s commands and acceptance rules are boxed in Appendix E; this section states what matters most.

Beyond robotics (future work). All measured evidence in this paper is the WorldEpisode robotics profile; the broader Universal Spatial State framing is proposed, not demonstrated outside robotics. 2 deterministic pilots only illustrate that the same vocabulary can express non-robotics drift. In a

game-engine collision pilot, a client loads a stale arena collision asset after an authoritative patch, which the asset-digest, state-revision, and role checks catch. In an autonomous-driving clock pilot, camera and lidar logs deserialize correctly, but a 50 ms undeclared clock offset at 15 m/s creates a 0.75 m fusion error against a 0.20 m tolerance, which the declared clock mapping corrects to 0.0 m. These are illustrative pilots, not production game-engine or AV benchmark results. A stronger paper should add at least one public non-robotics corpus and an independently maintained adapter.

Principal limitations. The headline holdout is confounded: task identity is part of the proxy-lineage key, so it measures a task–scene shift and cannot attribute the change to scene overlap alone. The MLP uses 5 optimization seeds and task one-hot features; crossed bootstrap intervals capture seed and held-out-episode variation, but 5 seeds remain a small training sample and the 0.25 nRMSE threshold is reported only as a secondary diagnostic rather than task success. ACT/Diffusion training and rollout evidence remain open; the pinned compatibility probe now completes 2/2 smoke-training paths after digest-pinned source-image materialization, but it produces no trained checkpoint, held-out policy metric, or rollout. The timing audit calibrates on 320 trajectories and evaluates 80 held-out trajectories, but it has no motor-effective timestamps and covers 1 robot type and 1 controller configuration. MuJoCo and Genesis still execute the same minimal servo model on 1 trajectory for the timing result. The separate contact-rich audit covers 2 scripted primitive tasks rather than an articulated robot or hardware reference; despite 1.000 task-outcome agreement, its 26.2° aggregate final-orientation error rules out equivalent physics. Conversion covers complete selected Parquet shards from 3 datasets, not their full corpora or source video payloads. Injected faults establish detection on known mutations, not natural prevalence or false-positive rates in deployment.

The implementation is also single-sourced. The clean-room reader is internal, the package is not yet an upstream LeRobot/Rerun integration, and no external compatible dataset is available. The 5-dataset natural-source audit mixes 2 active artifacts with 2 source-level metadata reviews and has no maintainer adjudication.

Societal impact. Fail-closed state checks may reduce unsafe or irreproducible robot evaluations and wasted training compute. Conversely, an incomplete contract can create false assurance, provenance fields can expose sensitive capture locations or operators, and content-addressed lineage can aid dataset linkage. Deployments should minimize or access-control provenance, retain native safety interlocks, and treat conformance as an audit aid rather than a safety certificate.

7 Conclusion

Robot-learning failures often arrive disguised as clean data: valid tensors, valid videos, valid scene files, and invalid state assumptions. WorldEpisode makes those assumptions explicit. It binds interaction data to immutable state revisions, gives persistent identity to entities, declares space/time/transition semantics, assigns roles to assets, records events and provenance, and reports conversion loss instead of hiding it.

In the reference experiments, a random split overlaps every task–scene proxy lineage; a task-confounded holdout increases mean episode nRMSE from 0.183 to 0.381 for the MLP and from 0.157 to 0.255 for deterministic temporal ridge. This diagnoses an invalid random split but does not isolate scene leakage. Freezing a 3-frame telemetry lag on 320 calibration trajectories reduces pooled RMSE from 4.845 to 1.934 over 80 held-out trajectories. This does not measure motor-effective latency. The validator catches every injected fault class. A preregistered 32-scenario MuJoCo/Genesis contact audit reaches 0.979 contact F1 and complete task-outcome agreement while exposing 26.2° final-orientation disagreement. The contribution is the auditable contract and its failure boundaries, not a claim of physical rollout improvement or equivalent physics.

The next experiments must disentangle task from physical scene lineage, run ACT/Diffusion and rollouts over matched seeds, instrument command and motor-effective timestamps on another controller, and anchor articulated contact replay to hardware observations. Until then, the paper makes no famous-benchmark inflation, simulator-independent physics, or mature-standard claim.

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468 Table 2 states the boundary of each empirical claim in Section 4.

Table 2: What the current evidence supports, and what it does not yet support.

Claim area	Current evidence	Boundary
Leakage	Public ArmnetBench LeRobot audit with 400 teleoperated reference episodes, an executable Torch BC probe, a measured temporal ridge state/action baseline, an ACT/Diffusion gate harness, and 4 compact digest-verified LeRobot state/action split packages totaling 241,470 frames	Lineage is a task-scene proxy and the holdout is task-disjoint; scene-only leakage is not identified. The LeRobot 0.6.0 preflight loads the representative package, but 0/2 policy paths complete a training step because split packages omit the required image/environment-state modality. After digest-pinned front-camera materialization, 2/2 policy paths complete a CUDA optimization step; held-out policy evaluation and rollouts are not run
Conversion	Complete pinned source Parquet shards from 3 public LeRobotDataset v3 datasets: 271 episodes and 43,601 paired state/action samples with exact tensor, index, and timestamp equality	Selected Parquet shards rather than full corpora; source video payloads are not converted or used for a throughput claim
Replay timing	Lag calibration on 320 SO-101 trajectories, 80 source-episode-disjoint evaluations across 8 tasks, tested same-trace MuJoCo and Genesis position-servo adapters, and scheduler conformance	Telemetry proxy with no motor-effective timestamps, 1 robot type, and 1 controller configuration; timing adapters use 1 trajectory
Contact-rich replay	Preregistered straight-push and parallel-jaw-capture protocol in MuJoCo and Genesis over 32 paired initial-state scenarios, with object, contact, grasp, pose, and outcome metrics	Scripted primitive actors, no articulated robot or hardware ground truth; final orientation differs by 26.2° overall, so equal physics is not claimed
Meta-simulator neutrality	Adapter contract matrix over MuJoCo, Isaac Sim, Genesis, and SAPIEN plus same-trace and contact-rich MuJoCo/Genesis evidence	Isaac and SAPIEN are not replay-tested in this paper, and equal physics is not claimed
Generalization beyond robotics	Deterministic game-engine collision-patch and autonomous-driving clock-domain pilots using the same invariant vocabulary	Not measured on production game engines, public AV datasets, or fleet logs
Validation	14 injected requirement faults, 2 independent hand-authored fixtures, and a pilot natural-source corpus over 5 public datasets	The 5-dataset count is met through active LeRobot artifacts plus source-level public benchmark metadata; dataset-specific diagnostic reports now cover all cases. Maintainer validation: <i>Not defined yet</i>
Binding retention	Versioned <code>uss-core-23</code> semantic projection checked by executable artifacts	Pilot projection; not a universal score of each storage format
Famous benchmark call-out	Source-level audit over Open X-Embodiment, DROID, BridgeData V2, LIBERO, and CALVIN, plus a targeted DROID subset rerun tool and a separate inflation-proof gate	Missing public controls are not measured score inflation; the proof gate currently has 100-episode DROID subset evidence but 0 measured inflation claims
Real-to-sim drift	Controlled action-contract and representation-role ablations where drifted contracts succeed in simulation and fail under deployment proxies	Deterministic proxy; not a RoboSnap/DROID-Sim rerun or physical hardware deployment
Adoption	Public schema, installable local preflight package, validator, fixtures, governance files, and an internal clean-room reader that does not import the reference SDK	No PyPI release, upstream LeRobot/Rerun integration, external independent implementation, or external dataset release yet
Scale architecture	Dataset manifest schema, scalable example, executable audit, and a generated 32,768-shard catalog benchmark describing 1,073,741,824 episodes for open/index latency, partition pruning, digest-cache behavior, and resolver routing	Catalog-side benchmark only; no billion episode rows, payload bytes, distributed storage, concurrent readers, or institutional federation are measured

469 B Experiment and Compute Provenance

470 The supplementary artifact includes a machine-readable experiment manifest at docs/experiments/
471 experiment_manifest/experiment_manifest.json. It joins 9 principal runs to pinned dataset
472 revisions, split and code digests, effective configuration, seed policy, output digests, and the remote
473 execution log. The measured environment provides an NVIDIA GB10 accelerator, 20 logical CPUs,
474 and 121.7 GiB RAM. The MLP run takes 46.69 s with 1510.7 MiB peak resident memory; the DROID
475 and controlled-suite processes take 0.62 s and 3.84 s. The conversion-scale process takes 48.24 s with
476 393.3 MiB peak resident memory; its 2 preliminary failed runs are retained with their nonzero exit
477 status and logs. The multi-trajectory timing audit takes 1.20 s with 247.6 MiB peak resident memory.
478 The logs also record storage and exit status. Temporal ridge shares the controlled-suite process and is
479 labeled with that accounting scope rather than assigned invented per-subtest compute.

480 The anonymous supplementary ZIP is built from an explicit allowlist with deterministic entry digests.
481 Source-control history and local environments are excluded; text payloads are anonymized before
482 hashing, binary payloads are scanned, and a separate audit checks extracted PDF text and metadata
483 plus every ZIP path and payload.

484 C Third-Party Assets

485 The generated inventory at docs/experiments/third_party_asset_audit/asset_audit.
486 json joins the experiment manifest to immutable dataset-card evidence. The 4 active datasets
487 are licensed as follows: ArmnetBench under Apache-2.0, SVLA SO-101 pick-place under Apache-
488 2.0, PushT under MIT, and the LeRobot DROID-100 mirror under MIT and CC-BY-4.0 (retaining
489 both the mirror and upstream DROID obligations).

490 The supplement contains 48 source-derived Parquet files in round-trip and split packages, covered
491 by 16 adjacent SOURCE_LICENSE.json records. It contains 0 source image or video files. Source-
492 derived rows retain their source expression; WorldEpisode-authored manifests, diagnostics, and
493 aggregate reports are CC0-1.0 unless stated otherwise. The generated THIRD_PARTY_ASSETS.md
494 records attributions, immutable card digests, installed software licenses, and context-only datasets.
495 Apache-2.0 and PushT MIT texts are distributed with the artifact.

496 The official NeurIPS 2026 style is pinned by digest. Because its distributed file contains no separate
497 SPDX or license statement, the inventory records its license as NOASSERTION and states the author-kit
498 submission use instead of inventing a license. This inventory is provenance documentation, not legal
499 advice.

500 D Pilot Binding Retention

Table 3: Semantic field retention in the versioned 23-field pilot projection. The reference binding retains 1.000 natively and is omitted. Sidecar recovery is by construction.

Binding	Native	With WorldEpisode
LeRobot v3	0.261	1.000
Rerun RRD	0.348	1.000
NCore	0.304	1.000
MCAP/ROS2	0.261	1.000
OpenUSD/SimReady	0.391	1.000
glTF Gaussian asset	0.174	0.304

501 E Open Gates: Commands and Acceptance Rules

Open result, not claimed: results not claimed in this release

502 The artifact keeps unfinished claims executable instead of hiding them. The current RFC release gate is: `python3 tools/release_readiness.py -strict-rfc`. The full-standard gate is: `python3 tools/release_readiness.py -strict-full`. The latter is expected to fail until the policy and benchmark gates boxed below are backed by committed reports and the machine-indexed maintainer, adoption, and simulator-equivalence gates are also closed. The same gates are indexed in `docs/experiments/open_reproduction_gates/open_reproduction_gates.json`.

Open result, not claimed: ACT/Diffusion and rollout impact

503 Result: *Not defined yet*. Rollout impact: *Not defined yet*. Scene-only leakage: *Not defined yet*. First replace the task-derived proxy with physical scene/source-capture IDs and construct task-stratified random and scene-lineage-disjoint splits. Then regenerate split packages and LeRobot job specs for those physical-lineage splits and run the generated ACT/Diffusion jobs inside a pinned LeRobot training environment. The committed front-camera materializer and successful compatibility smoke are prerequisites, not policy evidence.

Required artifacts: a proof that task support is identical across protocols and scene lineage is disjoint, policy checkpoint digests, train/eval configs with at least five seeds, offline action metrics, rollout traces or videos with content digests, and an updated policy-gate report.

Acceptance rule: both split protocols must preserve task support, at least one ACT or Diffusion Policy must report both protocols over multiple seeds, and at least one simulator or physical rollout must use the same manifests. Running the current task-confounded jobs does not close this gate.

Open result, not claimed: famous-benchmark score-inflation proof

504 Result: *Not defined yet*. A targeted DROID subset rerun can be attempted with `uv run -with pyarrow -with requests -with numpy python tools/famous_benchmark_policy_rerun.py -benchmark droid_100 -required`. The proof contract is then enforced with `python3 tools/benchmark_inflation_gate.py -required`.

Required artifacts: benchmark-specific WorldEpisode conversion, lineage/timing audit, faithful published-protocol policy rerun, paired corrected evaluation under the same metric, and a measured baseline-minus-corrected score drop.

Acceptance rule: the gate must contain at least one inflation-proof valid rerun report with `measured_inflation=true`. Source-level metadata gaps alone are evidence of missing controls, not evidence that published scores are inflated.

505 F Dataset-Scale Audit (DATASET.001)

506 To address production scale, we audit the dataset manifest in `examples/scalable-corpus.`
507 `worldepisode-dataset.json`. The audit validates 5 namespaces, 3 resolver schemes, 3 reg-
508 istries, 3 shards, 2 indexes, 1 append-only version, and 9 digest-addressed asset descriptors spanning
509 Hugging Face, S3, and OCI URIs with local mirrors. It also checks that the manifest includes a
510 world-lineage index, an asset-digest index, and a split-manifest shard. We then run a generated cata-
511 log benchmark with 32,768 trace shards describing 1,073,741,824 episodes. The committed report
512 records the measured time for parsing and indexing the generated JSON catalog, 8 partition-pruning
513 queries, digest-cache resolution, and resolver routing over all asset descriptors. This is catalog-side
514 evidence only, but it makes the scale claim executable: opening a corpus starts from the manifest and
515 indexes rather than from recursive storage discovery.

516 NeurIPS Paper Checklist

517 1. Claims

518 Question: Do the main claims made in the abstract and introduction accurately reflect the
519 paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract, Section 1, and Section 6 state the task–scene confound, one-trace replay scope, and open gates.

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Case 1 reports crossed seed–episode bootstrap intervals over 5 MLP seeds and episode intervals for deterministic ridge, but the 1-trajectory replay and exact conversion checks do not have repeated-run uncertainty estimates.

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Answer: [No]

Justification: Appendix B and the supplementary experiment manifest record host, NVIDIA GB10 accelerator, CPU count, RAM, storage, wall time, peak memory, and exit status for the principal remote runs. The answer remains No because failed preliminary work and total-project compute are not comprehensively accounted.

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